

PROGRAM MANAGEMENT GROUP CHAIRMAN'S
FACTUAL REPORT
(30 pages)



NATIONAL TRANSPORTATION SAFETY BOARD
OFFICE OF HIGHWAY SAFETY
WASHINGTON, D.C. 20594

PROGRAM MANAGEMENT GROUP CHAIRMAN'S FACTUAL REPORT

A. ACCIDENT

Accident: Tunnel Ceiling Collapse
Location: Eastbound Interstate 90 connector tunnel, about 1000 feet west of the Ted Williams Tunnel, Boston, Suffolk County, Massachusetts
Vehicle #1: 1991 Buick 4-door passenger car
Operator #1: Passenger Vehicle
Date: July 10, 2006
Time: 11:01 p.m. local time
NTSB #: HWY-06-FH-024

B. HUMAN PERFORMANCE GROUP

George W. Black, Jr., P.E.	(Chairman) <i>Senior Civil Engineer National Resource Specialist</i>
60 Forsythe Street, S.W.	Office of Highway Safety
Suite 3M25	404-562-2825
Atlanta, GA 30303	

C. ACCIDENT SUMMARY

About 11:01 p.m. local time on Monday, July 10, 2006, a 1991 Buick passenger car was traveling eastbound in the left lane of the Interstate 90 connector tunnel to the Ted Williams Tunnel in Boston, Massachusetts. The vehicle was occupied by a 46-year-old male driver and a 38-year-old female passenger. A section of the tunnel's ceiling structure, consisting of ten 8-foot by 12-foot concrete panels and associated hardware, collapsed and struck the Buick. The Ceiling was held in place by multiple stainless steel threaded-rod inserted into holes in the roof of the tunnel and secured with an epoxy adhesive. A post-crash inspection revealed that 20 of the anchors had pulled out from the

roof of the tunnel. These anchors were used to attach eight metal brackets that supported a portion of the ceiling structure. Initially, two of the concrete panels fell onto the passenger vehicle crushing the right side of the roof. Following this impact, the passenger car came to rest against the north wall of the tunnel, and additional panels came to rest on the vehicle. As a result, the female passenger occupying the front right seat was fatally injured and the male driver received minor injuries.

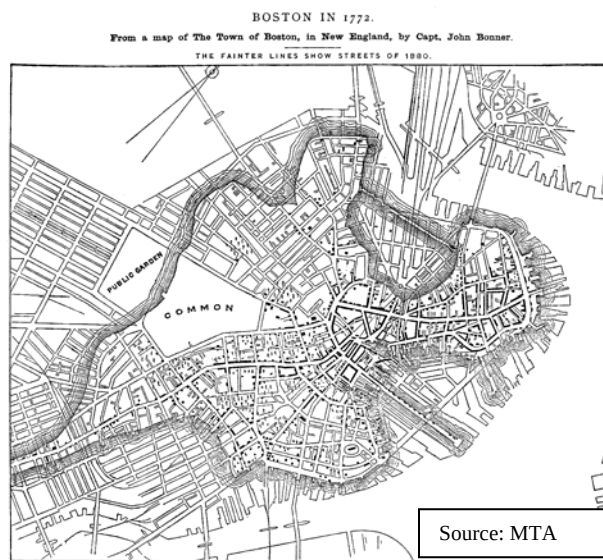
D. DETAILS OF THE INVESTIGATION

The Program Management factual investigation focused on the background of the Central Artery Tunnel Project, management of the project, including involved companies, and Federal and State oversight.

E. CENTRAL ARTERY TUNNEL (CA/T) PROJECT

E1. Background

Boston is one of the oldest cities in the United States having been founded in 1630. The City was originally located on a hilly islands and a peninsula in Massachusetts Bay, but expanded over the years to the surrounding mainland. Between 1630 and 1890, the city tripled its physical size by land reclamation, specifically by filling in marshes, mud flats, and gaps between wharves along the waterfront, a process of "cutting down the hills to fill the coves." The largest reclamation efforts took place during the 1800s. As the City grew further the irregular topography did not support a grid-type street network and an irregular network evolved.



Boston Street Map – 1772 and 1880

Motor vehicles replaced horse drawn vehicles and rail systems were built to service commuter traffic and Boston, Massachusetts developed severe traffic problem.

According to the Massachusetts Turnpike Authority's web site:

"To tell the story of the Central Artery/Tunnel Project, or the "Big Dig," it is necessary to go back to the road it replaced, the elevated John F. Fitzgerald Expressway - more commonly known as the Central Artery. Back in the 1940s - at the same time transportation officials were starting to envision the Massachusetts Turnpike - urban planners saw an increasingly tired-looking city of Boston choking on automobile traffic. The state Department of Public Works - the agency headed by the same man who would become the first chairman of the Massachusetts Turnpike Authority, William F. Callahan - had the novel idea of building a futuristic 1.5 mile "Highway in the Sky" through downtown Boston, with enough on- and off-ramps to service local destinations. The road was intended to work in concert with another highway, called the Inner Belt, that was to circle downtown Boston from the South End and Back Bay across the Charles River to Cambridge and Charlestown. Local traffic would use the elevated Central Artery, while through traffic would use the Inner Belt."

Construction on the Central Artery began in 1950, but the roadway was not without controversy. As late as 1954, the path of the final piece of construction was undecided, until the DPW decided to sink the remainder of the highway into a tunnel from Congress Street to Kneeland Street. This came to be known as the Dewey Square or South Station Tunnel. Even as it was being built from the Mystic Bridge south, the new artery, which displaced more than 20,000 people and demolished more than 1,000 structures, was seen as ugly and divisive of city neighborhoods. Its construction spurred citizen groups and others to successfully oppose the building of the Inner Belt. Central Artery construction concluded in 1959 with the opening of the Dewey Square Tunnel and the Southeast Expressway to Braintree. This new highway, part of Interstate 93, was designed for 75,000 vehicles per day. But from the outset, its 27 on- and off-ramps between Kneeland Street and Causeway Street, combined with a lack of merge and breakdown lanes, made for less than smooth traffic flow. Stall and crawl commuting was born. As early as the late 1960s, transportation planners began considering the next round of congestion-relieving highway construction, but it was not until the late 1980s when federal funding for a new, underground Central Artery and a new Boston Harbor tunnel connecting to Logan Airport was secured.

By the time Big Dig construction was getting into high gear in the mid-1990s, traffic on the elevated artery was 190,000 vehicles per day, with an accident rate four times the national average for urban interstates. Traffic was bumper-to-bumper for six to eight hours per day, with projections of traffic jams doubling by 2010. Plus, the elevated structure itself was decaying. After more than a decade of heavy construction, the Big Dig began paying dividends in 2003, with the completion of the I-90 connector to Logan International Airport in January. That new connection to the Ted Williams Tunnel meant

the Massachusetts Turnpike had reached its final, 138-mile length. Within months, the new I-93 underground artery and the Leonard P. Zakim Bunker Hill Bridge also opened to traffic.

The new Central Artery has the capacity to carry 250,000 vehicles per day. With the demolition of the old elevated highway, local traffic will travel along newly reconfigured surface streets, which will further facilitate traffic flow since far fewer vehicles will be using the underground artery for intra-city trips. Meanwhile, the Ted Williams Tunnel allows westbound traffic from the Massachusetts Turnpike to go directly to Logan Airport and the North Shore without having to use I-93. By carrying an estimated 90,000 vehicles per day, the Ted Williams Tunnel reduces the amount of traffic using the Sumner and Callahan Tunnels.”

The Williams Tunnel interchange in South Boston is at the center of a vital new development area for Boston, the seaport, with a new convention center and related projects. In addition to the new street and sidewalk grid being built in downtown Boston, new parks and open spaces will be constructed on the land that used to be in the shadows of a highway above.



Map of all BIG DIG projects

E2. Chronology of Boston Traffic and Safety Improvements

In 1983 the State, the City and the Federal Highway Administration FHWA initiated a transportation engineering study to evaluate alternative solutions to Boston’s

traffic needs. The results were two major roadway improvement components replacing the existing six-lane elevated highway with an eight-to-ten-lane underground expressway directly beneath the existing road, culminating at its northern limit in a 14-lane, two-bridge crossing the Charles River. After the underground highway opened to traffic, the elevated would be demolished and in its place would be open public space with some development.

The extension of I-90 (the Massachusetts Turnpike) from its former terminus south of downtown Boston through a tunnel beneath South Boston and Boston Harbor to Logan Airport. The first link in this new connection - the four-lane Ted Williams Tunnel (TWT) under the harbor was finished in December 1995. To construct these improvements (called the Central Artery/Tunnel) proved to be one of the largest, most technically difficult and environmentally challenging infrastructure projects ever undertaken in the United States. The project spans 7.8 miles of highway, 161 lanes miles in all with about half in tunnels. The larger of the two Charles River bridges, a ten-lane cable-stayed hybrid bridge, is the widest ever built and the first to use an asymmetrical design. The project also includes four major highway interchanges to connect the new roadways with the existing regional highway system.

At Logan Airport, an interchange carries traffic between I-90 and Route 1A as well as onto the airport road system. In South Boston, a mostly underground interchange carries traffic between I-90 and the waterfront and convention center area. At the northern limit of the project, a new interchange connects I-93 north of the Charles River to the Tobin Bridge, Storrow Drive, and the new underground highway.

At the southern end of the underground highway, the interchange between I-90 and I-93 is was completely rebuilt on six levels -- two subterranean -- to connect with the underground Central Artery and the Turnpike extension through South Boston. The interchange includes a total of 28 routes, including High Occupancy Vehicle (HOV) lanes, and channels traffic to and from Logan Airport to the east. A fifth interchange, at Massachusetts Avenue on I-93, was rebuilt in the project.

The Massachusetts Turnpike Authority (MTA) believes that The Central Artery/Tunnel Project is public works on a scale comparable to some of the great projects of the last century -- the Panama Canal, the English Channel Tunnel (the "Chunnel"), the Trans-Alaska Pipeline. Each of these projects presented unique challenges: The Panama Canal confronted earth-slides, malaria, yellow fever, and Central American jungles. The Chunnel was dug from either end, 31 miles apart, meeting at a precise point under the channel floor. The Alaska Pipeline contended with vast distances, freezing temperatures, and major environmental concerns.

The Central Artery/Tunnel Project is owned and managed by the Massachusetts Turnpike Authority (MTA), and is part of the Metropolitan Highway System (MHS). The expected final cost of the program is over 14 billion dollars.

E3. Program Management

The MTA's web site states that:

"The Massachusetts Turnpike Authority was created by an act of the Massachusetts Legislature in 1952 and did not receive state or federal tax revenue. The roadway, including the Boston Extension in the Metropolitan Highway System (MHS) and the two tunnels, operates on toll revenue, supplemented with revenue from leasing, development of land and air rights, and advertising." In 1997 Massachusetts law assigned the Massachusetts Turnpike Authority the responsibility of overseeing the Central Artery/Ted Williams Tunnel (CA/T) Project. Upon completion all CA/T roadways will become part of the Massachusetts Turnpike Authority's MHS. The Massachusetts Highway Department (MHD) is responsible for the design construction and maintenance of all state highways in Massachusetts other than those of the MTA. With regard to the CA/TA, according to a 1997 agreement between the MTA and the MHD, the MHD is responsible for Funding, Contract Approval¹ and oversight of those projects. MHD is also responsible for funding, coordination and oversight with the Federal Highway Administration (FHWA). MTA and MHD would be responsible for:

- Ultimate decision maker for CA/T Project issues, including those affecting cost, claims and schedules.
- Assure adequate funding to implement the Project (prepared finance plans).
- Procurement of professional consultants, (Program Manager and SDCs).
- Awards and executes all construction contracts.
- Oversight and management of Program Manager.
- Operations and maintenance responsibility."

E3.1. PROGRAM MANAGER

The State realized early on that the program was too large to be managed on a day to day basis by the MTA and/or MHD. It was decided that consultants would be engaged to furnish most engineering services and day-to-day program and individual project management. This means of managing large construction programs is often used by State and local governments. It is impractical to employ permanent employees for one program and then have to either dismiss them or try to find other work for them after the program is completed. The State solicited proposals for a single program management contract that would provide the following services²:

¹ For example, all of the contracts between design consultants (SDC) and project managers were between the MHD and those firms and not the MTA.

² There are numerous other smaller service items in the contract.

- Preliminary Design
- Final Design Coordination and Review
- Construction Coordination and Monitoring by inspectors and resident engineers
- Preparing Cost Estimates and Reporting to MHD/MTA on the Project's overall cost and schedule
- Right of way and other property acquisition
- Computer aided design record keeping
- Utility engineering and coordination
- Internal Quality Assurance
- Construction Safety Programs
- Owner's authorized representatives for Construction Contract administration
- Standard Drawings for the program
- Geotechnical engineering services
- Environmental services
- Engineering materials testing

In 1985, The State entered into an agreement with Bechtel/Parsons Brinckerhoff—a joint venture formed between Bechtel Corporation and Parsons Brinckerhoff Quade & Douglas—to prepare a preliminary project management plan. Later the same year, the department issued a 1-year contract to Bechtel/Parsons Brinckerhoff (B/PB) to develop a comprehensive work plan for managing design consultants and reviewing their work. This contract was followed by a number of limited-term contracts with B/PB, called “work programs,” that extended through the life of the project..

The final design of each project section was the responsibility of a design engineering firm or firms³ through the award of Section Design Consultant (SDC) contracts with the following responsibilities:

- Develop and stamp⁴ detailed design sufficient for Construction Contractors to bid and build the various Section Contracts of the Project.
- Provide construction phase services such as review of shop drawings, submittals, and responses to a Construction Contractor's Request for Information (RFI).

³ Some projects were large enough to support “joint-ventures” by two or more individual engineering firms.

⁴ “Stamp” means having a Massachusetts registered professional engineer approve the plans by affixing their seal.

E3.2. BECHTEL/PARSONS BRINKERHOFF

Bechtel's and Parsons' web sites and other B/PB information provided the following corporate description:

"Founded in 1898, Bechtel is one of the world's premier engineering, construction, and project management companies. Our 40,000 employees are teamed with customers, partners, and suppliers on a wide range of projects in nearly 46 countries." Bechtel has completed more than 22,000 projects in 140 countries, including Hoover Dam, the Channel Tunnel, Hong Kong International Airport, the San Francisco Bay Area Rapid Transit (BART) system, the reconstruction of Kuwait's oil fields after the Gulf War, Jubail industrial city, and the Alma aluminum smelter."

E3.3. PARSONS BRINKERHOFF

"The past years have seen PB play significant roles on thousands of projects, including some of the most notable infrastructure endeavors of the 20th century. It is these projects that tell the complete story of our heritage: The legacy of our impact around the world. It is a legacy we continue, as we help today's clients serve the public in nearly 80 countries around the world. PB can manage the full range of activities necessary to keep a complex multidisciplinary project—often worth billions of dollars—firmly on track. Program management is more comprehensive than project management, design management or construction management. A program manager often oversees multiple project managers that are executing various aspects of the program of work."

The two firms staffed the positions needed to manage the State's programs and operated a materials laboratory that tested some of the materials that went into the projects in accordance with the State's Materials Manual and AASHTO guidelines.

F. BOSTON CENTRAL ARTERY/TUNNEL PROJECT PROGRAM MANAGEMENT

The Boston Central Artery/Tunnel (CA/T) project was planned, designed, and constructed over a period of 20 years from 1985 through 2006. The CA/T project adopted a conventional design-bid-build approach to the various project sections and was set up with layers of public oversight and checks and balances. In its role as management consultant, B/PB was accountable initially to MHD. In its role as management consultant, after 1992 its responsibility was to MHD and, after 1997, to the Massachusetts Turnpike Authority (MTA). MHD/MTA, in the role of owner, was responsible for oversight of B/PB and all significant decisions concerning projects to be built, schedules and budgets.

B/PB assisted in the decision-making process by developing alternatives and presenting recommendations to the owner on appropriate solutions. Subject to the

owner's oversight, B/PB functioned as the owner's representative in specified areas concerning day-to-day administration of DPW/MHD's design and construction contracts.

As part of its oversight role, DPW/MHD/MTA engineers and project management staffs reviewed and evaluated B/PB's work. The Federal Highway Administration (FHWA) also participated in reviews and oversight of almost all major B/PB recommendations and MHD/MTA decisions. This FHWA oversight applied to all projects on the Federal Interstate highway system.

As management consultant, B/PB prepared the concept studies and preliminary design basis for the CA/T, representing about 25% of the total design effort. DPW/MHD contracted with a number of section design consultants (SDCs) selected by section specific panels of public employees and approved by the DPW/MHD's Architect and Engineering Board. The selected SDC reviewed the preliminary design, prepared and sealed⁵ a final detailed design. Each SDC was responsible for the detailed final design, as well as for construction specifications. B/PB's review of the SDC's design focused on conformance with project scope, DPW/MHD and project standards, design criteria and interfacing requirements.

In June of 1997 the State issued a "Central Artery/Ted Williams Tunnel Project Management Agreement" that changed management, "ownership" and other responsibilities for the CA/T projects. This agreement also substantially altered B/PB's role in by combining key B/PB personnel with those of the MTA in an Integrated Project Organization (IPO). MTA's stated goal was to streamline project management and improve cost effectiveness. The process also allowed for a transition from B/PB personnel to the State as the program was completed.

After MTA assumed overall management in 1997, MHD remained responsible for contracting with SDCs and construction contractors. With B/PB's assistance, MHD advertised the SDC's final design and specifications, typically receiving bids from three or more construction contractors. The contractor selected and contracted by MHD was then responsible for constructing the section in accordance with the contract design and specifications and its own quality control (QC) program – and certifying that it had done so.

Prior to commencement of the work, B/PB reviewed the contractor's project-specific quality control program for conformance to standards set out in the contract. During construction, the contractor performed the QC, including as necessary the provision of material certificates and proof testing by an independent agency approved by MHD. Certain materials used in the contractor's work were also tested by the CA/T project's construction laboratory, which was managed by B/PB, certified by the American Association of State Highway and Transportation Officials (AASHTO), and independently assessed by MHD.

⁵ The term "sealed" in this usage means that the plans were certified by a Massachusetts registered professional engineer employed by the SDC.

As part of the CA/T project's quality assurance (QA) program, B/PB (and after 1998, the IPO) performed quality assurance checks of the contractor's work for general compliance with contractual requirements and approved submittals. It oversaw the contractor's QC program. Responsibility for control of quality belonged to the contractor, however, in accordance with their contract with DPW/MHD and general industry practice.

Contractor questions or requests for information or clarification relating to the SDC's final design were typically referred to the SDC for resolution rather than B/BP. Accordingly; the SDC remained actively engaged throughout the construction phase, as the engineer of record. B/PB facilitated the resolution of issues between or among contractors, SDCs, and DPW/MHD/MTA. In addition, significant issues and challenges that arose during the design and construction of the CA/T project were typically brought to the attention of the Project Interface Committee, comprised of the MTA/MHD, FHWA and B/PB, which met weekly. The expected final amount of the fee paid to B/PB is over two billion dollars.

G. FEDERAL HIGHWAY ADMINISTRATION AND STATE OVERSIGHT

The following is the oversight agreement between the State and the FHWA. Federal Highway Administration Oversight Agreement dated 3/8/1999. When the CA/T program is completed the Federal Highway Administration will have provided \$8.549 billion for the Big Dig Program. Essentially all of the individual projects were a part of the Interstate System and received "Full FHWA Oversight" as required by federal regulations and the agreement between the State and FHWA and the following agreements.

*"Massachusetts Highway Department
and
Federal Highway Administration - Massachusetts Division*

In recognition of the long standing partnership between the Massachusetts Highway Department and the Massachusetts Division of the Federal Highway Administration and in furtherance of Section 106 of Title 23, "Project Approval and Oversight", as amended by Section 1305 of TEA- 21, this agreement has been jointly developed to assign appropriate oversight responsibilities for various programs while retaining the historical value-added aspects of the partnership for Program Delivery. The "responsibilities" involved with this agreement are limited to Design, Plans, Specifications, Estimates, Right-of-way Certifications (part of PS&E), Contract Awards (PS&E)⁶, Contracts and awards of Federal-aid Projects as categorized below.

⁶ The FHWA's policies for construction authorization and plan, specification and estimate (PS&E) approval provide for: The FHWA's review of the PS&E documents for certain projects (as described in the Division

Projects on the Interstate System – FHWA full oversight responsibilities are to be maintained.

Projects on the NHS (but off the Interstate System) - MHD will assume the Project responsibilities for all Projects with an estimated Construction Cost less than \$5 million. Projects in excess of this amount will continue to have PHWA Oversight.

Projects off the NHS - MHD will assume the responsibilities noted for all projects except those in excess of \$10 million which will have FHWA oversight responsibilities.

All general Title 23 requirements that apply to all projects, such as metropolitan and statewide planning, procurement of services or contracts, disadvantaged business enterprises, wage rates, etc., continue to apply regardless of oversight responsibility noted above.

In the event that unique circumstances arise, or for joint program/process reviews, FHWA involvement with any Federal-aid project is recognized as a-required element with appropriate coordination with MHD.

FHWA Technical Assistance will continue to be provided as requested on any project, and FHWA involvement will continue in all existing "Committees" and continue to be available in any specific MHD requested Task Force or problem solving groups.

Signed:

Commissioner, Massachusetts Highway Department

Division Administrator, Federal Highway Administration"

On December 1, 1999, and subsequently on January 5, 1999 and November 17, 2000 the FHWA and issued an updated "Construction Monitoring Program Memoranda to accommodate the changes in State management of the Big Dig Program. It

Office's Stewardship Agreement), and certifications by the contracting agency that clearances have been obtained (or necessary arrangements have been made) so as not to delay the construction contractor by delays due to inadequate coordination for right-of-way, utility relocations, railroad relocation, NEPA environmental clearances, and other pre-construction coordination programs.

incorporates the changing relationships between the MTA, MHD and B/PB and the Integrated Project Organization (IPO). The following is the 11/17/200 Memorandum:

“Subject: Construction Monitoring Program FY 01 November 17, 2000
From: FHWA Divisional Director
Project Delivery Team Leader
In Reply Refer To:
To: All Area Engineers

Our Construction Monitoring Program of Federal-aid contracts has been developed to assess procedures and controls for assuring that contracts are completed in reasonably close conformity with the approved plans, specifications, and approved changes, as well as for the monitoring and evaluation of progress. (The Project Oversight Agreement between MHD and FHWA as executed on March 8 and 1999 specifies the extent of FHWA oversight for Federal-aid contracts.) The Program is also designed to provide coordination with MHD, MTA, B/PB for CA/T) representatives so as to assure we maintain a working knowledge of the construction contracts, especially in regard to developing issues and/or potential contract changes. Our early involvement in potential contract changes is enhanced by our continued participation at the Area and Statewide Districts meetings. The relationship with MHD or MTA and B/PB staff will also allow FHWA to provide timely advice on technical or procedural issues. This working relationship will continue to be fostered by our participation at the Area Managers Meetings, and on the Statewide Program by our routine meetings with MHD District staff.

Appropriate inspection strategies for each construction project will be coordinated with the Project Delivery Team Leader and will be guided by the following:

Initial inspection - will be conducted on each contract when physical activities commence (usually within 3 months of N-TP).

Project Inspection - will be conducted generally on a monthly basis for all major contracts, defined as a Contract with a value exceeding \$5 million. Other contracts will receive coverage commensurate with activities underway. The areas to be reviewed on individual inspections will be guided by activities in progress and known potential problem areas. Areas that should be periodically assessed are; (1) Construction Staffing -- in order to assure the appropriate project inspection staffing needs, and (2) Traffic Control in Work Zones -- as an activity which will support our safety performance Goal of reducing accidents.”

On 22 June 2000 the FHWA, the Executive Office of Transportation, MHD and the MTA entered into an agreement entitled the “CENTRAL ARTERY/TED WILLIAMS TUNNEL PROJECT PARTNERSHIP AGREEMENT”. The apparent intent of this document was to manage and control the cost increases that the program had experienced.

***“CENTRAL ARTERY/TED WILLIAMS TUNNEL
PROJECT PARTNERSHIP AGREEMENT*”**

This agreement is entered into by and between the Federal Highway Administration (“FHWA”), the Commonwealth of Massachusetts (“Commonwealth”) acting by and through the Executive Office of Transportation and Construction (“EOTC”) and the Massachusetts Highway Department (“MHD”), and the Massachusetts Turnpike Authority (“MTA”).

WHEREAS, pursuant to Massachusetts General Laws Chapters 16 and 81A, MHD and MTA, effective July 1, 1997, entered into the Central Artery/Ted Williams Tunnel Project Management Agreement, under which, among other things, MTA agreed to assume management of the Central Artery/Ted Williams Tunnel Project (“CA/T Project”) on behalf of MHD, and

WHEREAS, MTA is responsible for management and supervision of the CA/T Project and is currently using an Integrated Project Organization (“IPO”) through the engineering and management services of the joint venture of Bechtel Corporation and Parsons Brinkerhoff, Quade, and Douglas, Inc. (“B/PB”) and,

WHEREAS, MHD, acting through EOTC, is the recipient of Federal Highway Funds for the Commonwealth, and as such, oversees the obligation and distribution of federal funds to the CA/T Project, and performs audit functions consistent with said funding and,

WHEREAS, B/PB reports to and receives direction and supervision from the Commonwealth acting through MTA and in turn B/PB implements that direction throughout designated phases of the CA/T Project and, WHEREAS, FHWA seeks to clarify its oversight role in implementing the CA/T Project and to establish new reporting and verification procedures necessary to ensure the Commonwealth’s accurate and full financial disclosures regarding the CA/T Project and,

WHEREAS, this agreement will be deemed to supplement the existing MHD/FHWA Massachusetts Division Office Project Oversight Agreement as it relates to the CA/T Project,

NOW IT IS THEREFORE AGREED, by and between the parties to this agreement:

FUNDING

1. Notwithstanding the Commonwealth’s use of advance construction (“AC”) authority the maximum level of Federal-aid highway obligation authority (“OA”) and funding available for the CA/T Project shall not exceed \$8.549 billion.

2. In recognition of the CA/T Project's funding levels previously established in accepted finance plans and updates, and the Commonwealth's commitment to provide funds necessary to cover the recently disclosed cost increases, the maximum amount of new net advance construction authorizations on the CA/T Project shall not exceed \$222 million. This will provide an AC balance to improve cash flow management and allow the CA/T Project to recover funds resulting from underruns on existing contracts. In no case shall AC conversions for the CA/T Project be allowed which bring the OA total above the \$8.549 billion dollar cap.

RESPONSIBILITIES

3. To the extent allowable by law, all state and private entities associated with the CA/T Project shall provide any and all information and records to external monitoring agencies and shall provide timely access to records and persons employed on the CA/T Project. MHD and MTA shall be held accountable for compliance with this provision and a failure to provide timely and full access may be considered a violation of 49 CFR 18.42(e) with potential imposition of sanctions including but not limited to withdrawal of reimbursement and further availability of federal-aid funding. MHD will support the MTA's efforts to provide timely and full access of records.

4. The MHD, MTA and Massachusetts's Executive Office for Administration and Finance (A&F) shall provide to the FHWA all data with respect to independent audits of the CA/T Project including but not limited to O'Brien Kreitzberg and Deloitte Touche audits, and failure to provide such data will similarly be viewed as a potential violation of the regulations subject to potential imposition of sanctions. The FHWA will independently review these audits as well as those performed by other entities and coordinate needed action with appropriate parties.

5. The CA/T Project shall perform a total project forecast for the remaining years, in conjunction with the annual finance plan, beginning with the October 1, 2000 submission. The results of these reviews will be included in the project monthly management report ("PMM") and the annual finance plan updates.

6. The CA/T Project shall assure that the data contained in the PMM is modified to show potential cost exposures. The PMM shall also include such items as:

- a. anticipated cost exposures in design and/or construction activities
- b. projected labor rate increase
- c. anticipated petroleum price increases or decreases
- d. expected increases in operational costs such as insurance premiums, consultant support services, and materials, and

FUNDING

1. Notwithstanding the Commonwealth's use of advance construction ("AC") authority the maximum level of Federal-aid highway obligation authority ("OA") and funding available for the CA/T Project shall not exceed \$8.549 billion.

2. In recognition of the CA/T Project's funding levels previously established in accepted finance plans and updates, and the Commonwealth's commitment to provide funds necessary to cover the recently disclosed cost increases, the maximum amount of new net advance construction authorizations on the CA/T Project shall not exceed \$222 million. This will provide an AC balance to improve cash flow management and allow the CA/T Project to recover funds resulting from underruns on existing contracts. In no case shall AC conversions for the CA/T Project be allowed which bring the OA total above the \$8.549 billion dollar cap.

RESPONSIBILITIES

3. To the extent allowable by law, all state and private entities associated with the CA/T Project shall provide any and all information and records to external monitoring agencies and shall provide timely access to records and persons employed on the CA/T Project. MHD and MTA shall be held accountable for compliance with this provision and a failure to provide timely and full access may be considered a violation of 49 CFR 18.42(e) with potential imposition of sanctions including but not limited to withdrawal of reimbursement and further availability of federal-aid funding. MHD will support the MTA's efforts to provide timely and full access of records.

4. The MHD, MTA and Massachusetts's Executive Office for Administration and Finance (A&F) shall provide to the FHWA all data with respect to independent audits of the CA/T Project including but not limited to O'Brien Kreitzberg and Deloitte Touche audits, and failure to provide such data will similarly be viewed as a potential violation of the regulations subject to potential imposition of sanctions. The FHWA will independently review these audits as well as those performed by other entities and coordinate needed action with appropriate parties.

5. The CA/T Project shall perform a total project forecast for the remaining years, in conjunction with the annual finance plan, beginning with the October 1, 2000 submission. The results of these reviews will be included in the project monthly management report ("PMM") and the annual finance plan updates.

6. The CA/T Project shall assure that the data contained in the PMM is modified to show potential cost exposures. The PMM shall also include such items as:

- a. anticipated cost exposures in design and/or construction activities
- b. projected labor rate increase
- c. anticipated petroleum price increases or decreases
- d. expected increases in operational costs such as insurance premiums, consultant support services, and materials, and
- e. estimated and settled Right of Way (ROW) and construction claims.

7. The CA/T Project shall document projected delays to schedule milestones, identify the causes for these delays, and provide measures under consideration to remedy these

delays. This information shall be provided on a timely basis and in a manner enabling FHWA to comment on these matters and coordinate in an appropriate way.

8. EOTC and MHD shall reach agreement in principle with the MARPA respecting the terms of a balanced statewide program. The meeting of this balanced statewide program, and concurrence of MARPA, will be a formal condition for granting STIP approval.

9. The Secretary of EOTC and the Commissioner of MHD shall certify that each CA/T Project finance plan and update is consistent with the balanced statewide program and the approved STIP. This certification shall be required as a condition of FHWA's acceptance of the finance plan and updates.

10. The CA/T Project shall ensure that the potential forecast for all CA/T Project elements be a best estimate of the completion cost, be maintained on a current basis for all CA/T Project elements, and be accurate and complete and unconstrained by policy directives regarding costs of the CA/T Project.

11. The CA/T Project shall submit an overall Budget vs. Potential Forecast Variance Report on a semiannual basis to the FHWA beginning October 1, 2000. This report shall contain an explanation of any and all variances, by CA/T Project element, segregated into the following categories:

a. the components of the reported variances that are deemed to be firm to the point of requiring a corresponding revision to the budget of the affected CA/T Project element.

These include but are not limited to, the value of actual contract awards (or executed change orders), approved scope changes to be incorporated during design, and expected settlement amounts for asserted construction claims.

b. the components of the reported variances that are deemed by the CA/T Project management to be subject to further adjustment by future management corrective action, or other alternative remedies.

c. the components of the reported variances that are deemed as being speculative in nature. These include speculative forecast variances that are difficult to quantify and price but which could have a positive or adverse effect on the future cost of the CA/T Project and/or the statewide program.

The FHWA, the EOTC, MHD and the MTA have caused this Agreement to be executed by their authorized officers as a sealed instrument effective as of 22 June 2000.

FEDERAL HIGHWAY ADMINISTRATION EXECUTIVE OFFICE OF
TRANSPORTATION AND CONSTRUCTION

By: signed

Division Administrator Secretary

MASSACHUSETTS TURNPIKE, MASSACHUSETTS HIGHWAY
AUTHORITY DEPARTMENT

By: signed _____

Chairman

By: signed _____

Commissioner

Approved as to form:

By: signed _____

Acting General Counsel

By: signed _____

Associate Commissioner

By: signed _____

Associate Commissioner

NTSB staff members met with the FHWA Divisional office in Cambridge, MA. The staff was very knowledgeable about the State's construction program and supplied numerous documents to the Board. During the CA/T additional staff was provided by the FHWA from Washington, resource centers and other divisions to assist in oversight of the program.

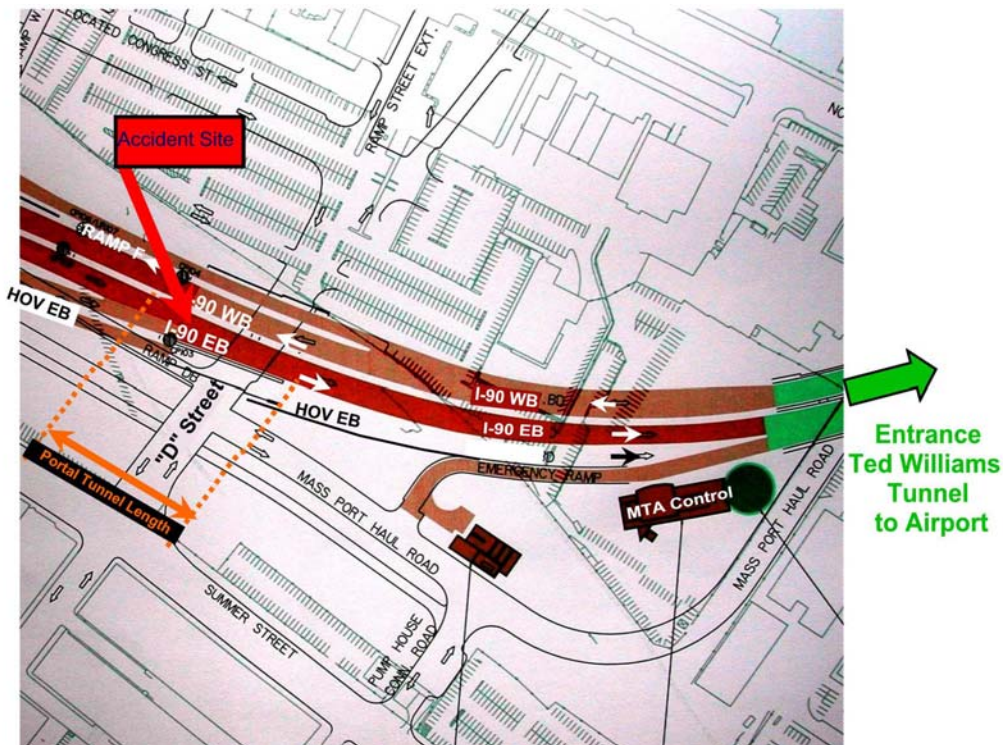
H. ACCIDENT PROJECT – C09B2

The site of the accident was constructed as a part of the "Ted Williams Tunnel" (TWT) project which was opened to traffic in 1995. The TWT was designed to connect the Logan Airport with the southern area of Boston, I-90 and I-93. The accident site was called the "Portal Tunnels", Seaport Access Tunnels, or the "West Tunnels"⁷. Part of this contract was to construct an extension of "D" Street to facilitate area traffic movement until the I-90 Central Artery Tunnels could be completed.

The individual tunnels were designated "HOV-East Bound, I-90 East Bound and I-90 West Bound. (The accident occurred in the I-90 East Bound tunnel.) The total length of the tunnels was about 250 feet. At the time of its construction these tunnels were intended to be "finished"⁸ in the same manner as the TWT. This would have included the light weight plenum ceiling panels secured with epoxy anchor bolts as in the TWT.

⁷ Over this time period, different organizations gave several names to these tunnels.

⁸ A tunnel "finish" project installs all of the items necessary for operation of the tunnel. Such items as ventilation, electrical connections, lighting are included in these contracts.



Plan View of Accident Tunnel Area

The Section Design Consultant for the Portal Tunnels and their connection to the TWT was HDR, Inc. Their design of the portal tunnel ceiling assumed the same design as philosophy as used in the TWT – light weight ceiling panels suspended by adhesive anchors. This design was not installed because the ventilation system designs for the I-90 tunnels were not complete.

With respect to the I-90 central tunnel connector from I-93 to the east end of the accident tunnels, B/PB completed the preliminary design, in 1992. The MHD then selected and contracted with Gannett Fleming as a Section Design Consultant (SDC) in 1993 and added the tunnel finishes for the I-90 connector portal tunnels, including the ceiling re-design, to this contract in 1997. MHD awarded Modern Continental the construction contract in 1998.

Based on experience with the ceiling system designed and installed in the TWT, B/PB and the State and the FHWA decided to seek a different design for its ceiling installations. The TWT ceiling consisted of light relatively light weight panels that were secured to the tunnel's top by adhesive anchor bolts. They have fairings that produced a smooth transition with the walls. The FHWA, State and B/PB felt that this design was too expensive procure, install (with epoxy anchor bolts) and maintain. Replacement panels were expensive and the design did not have a uniform sizes for all panels. The FHWA, the State and B/PB felt that these features would significantly increase the maintenance costs over the life of the ceiling.

The cost of the over all program had increased significantly for the initial estimates. All of the involved agencies were searching for cost control alternatives on all projects remaining in the program. The Central Artery including the accident tunnels.

The result of this evaluation was a new design first used in the I-93 tunnels north of the interchange of I-93 with I-90. However, these tunnels had constructed with exposed ceiling beams providing a location for fittings to attach the panels above the roadway. The individual panels weighed 50 pounds per square foot compared to the 20 pounds per square foot for the TWT panels. Because of the extra weight, the support system would also be heavier.

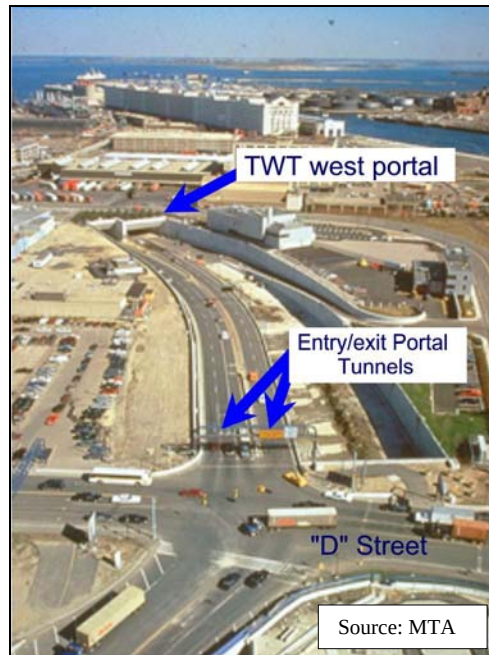
Thus the accident area ceiling for I-90 between I-93 including the Portal Tunnels was designed to utilize the heavier less expensive ceiling panels. A different system of attaching the ceiling panels to the top of the tunnel was also specified. During construction the I-93 tunnels had embedded transverse steel channels cast into the Portland cement concrete tunnels' tops. This allowed for ceiling panels to be suspended by steel rods connecting the embedded transverse steel channels in the top of the tunnel. This suspension design was included in the I-93 tunnel⁹ by the finishes consultant Gannett Fleming Engineers.

The "finishes" contract for the I-93 tunnels included the "portal" or "west" tunnels which were built earlier did not have the embedded transverse steel channels devices installed. Since the portal tunnels had been originally intended to have adhesive anchor bolt suspended ceiling panels, the ceiling suspension system was designed by the SDC Gannett Fleming and reviewed by B/PB.

The tunnel finishes contract (Project C09B2) was awarded to Modern Continental Inc. The contractor was given a notice to proceed on 16 November 1998. This project included the tunnels between I-93 on the west to the east end of the Portal Tunnels where the accident occurred.

The TWT was opened to commercial traffic in 1995 and the portal tunnels were utilized to connect the existing street network to the TWT. This traffic pattern was utilized until the opening of the I-90 artery tunnels to I-93 in 2003.

⁹ The design did not function exactly as it was intended to. Many of the embedded steel features were too deep to reach without possible damage to the structural integrity of the top of the tunnel. Others were found to be in the wrong location. The result was that about 10-15% were unusable and epoxy anchor bolts had to be substituted.



Accident Tunnel area looking east

H1. Project Management for Tunnel Finishes contract (Project C09B2)

MTA produced an organizational chart for the accident project C09B2.

H2. Quality Control/Quality Assurance

The American Association of State Highway and Transportation Organizations (AASHTO) and the Federal Highway Administration FHWA established Quality Control requirements for highway construction projects in their publications “AASHTO – “Implementation Manual for Quality Assurance, February 1996”, “AASHTO Construction Manual for Highway Construction 1990” and FHWA’s “Quality Assurance Procedures for Construction”, published on June 29, 1995 as 23 CFR 637.”

These documents define quality assurance as “All those planned and systematic actions necessary to provide adequate confidence that a product or service will satisfy given requirements for quality.” The Specification goes on to state that, “The term Quality Control and Quality Assurance (QC/QA) has often been used synonymously with Quality Assurance. The above definition considers Quality Assurance to be an all encompassing concept which includes quality control (QC), acceptance, and independent assurance (IA). Therefore, Quality Assurance is the proper term and is used in this manual.”

Under the heading of “Engineering Personnel” the AASHTO Specification states, “Many Agency engineering personnel have been involved in construction management for many years. They may believe that there is no need to change and that you are trying to “fix something that isn’t broken.” Some may interpret the Quality Assurance program

as a form of automation and believe that they are being stripped of responsibilities. They may feel that their only responsibilities are to "observe and record" test results and when the work is done, just apply the proper pay adjustment. The inspectors must realize that their responsibility has not changed and that inspection is still a critical element of acceptance. If a Contractor is producing unacceptable work, the Contractor must be notified as soon as possible."

It goes on to say that, "The Contractors must understand that QC is their responsibility, and "Acceptance" is the Agency's responsibility. It is important to realize that some of the test results will not be identical; therefore, an Agency must address these instances in their specifications. Contractors should establish correlation factors between their QC tests and the Agencies acceptance testing.

Basically the QC/QA system developed by the State DOTs, AASHTO and the FHWA has dictated a process where the Contractor produces certifications from suppliers that a material meets the State's specifications. Then the State's materials testing group acquire random samples and subject them to testing in the State's certified laboratory or a certified private consultant's lab. The process also tests primarily structural materials for a project. This would include pavement materials (asphalt, Portland cement, pavement and structural concrete aggregate, structural steel, and admixtures to pavements. Board investigators asked B/PB if its laboratories had conducted a "pull test" on any of the epoxy anchor bolts in the tunnels where they were installed or tested the epoxy material. They had not, but they did test the steel anchor bolts and other structural steel components. When asked, other state DOTs described a similar QC/QA process. However, five state DOT materials engineers were interviewed on QC/QA and all described the use of a "risk based" process that required their independent testing of potentially critical material components. When asked if they would have considered epoxy adhesive anchor bolts securing concrete components over an active roadway, all stated that this design would have justified their risk assessment testing process.

This lack of adequate QC/QA was the result the formulation of a QC/QA policy by AASHTO in the early 1990's. This model QC/QA program placed responsibility for **Quality Control** on the project construction contractor and their materials testing subcontractor. The construction manager (a state or its PM contractor) then had the responsibility for **Quality Assurance** through monitoring the QC of the contractor and its material testing consultant. (Candid and confidential comments by personnel associated with this state that the two driving forces behind it were tort liability and the personnel costs of maintaining competent state testing laboratories. Not all states changed to the process. AASHTO did maintain (along with ASTM) specifications for many materials but did not test samples of actual materials to be used if they were "certified" to meet the AASHTO/ASTM specifications.

In the case of the accident project, the steel anchor bolts and the tunnel's Portland cement concrete were tested by the CA/T project's construction laboratory, which was managed by B/PB, but never tested the epoxy. The anchor bolts' epoxy was tested only for its performance in high temperatures resulting from any possible tunnel fires.

There is also a risk management issue in this accident. It appears that the B/PB the state and the contractors spent considerable time on the Ted Williams Tunnel to assure that the epoxy anchor based suspension system were reliable. But on the accident tunnel they adapted a less expensive heavier ceiling module system a short segment of tunnel without considering the risks even though some B/PB construction supervision personnel on the project documented their concerns about the process.

At the request of the Safety Board, AASHTO conducted a survey of its members and their QA/QC practices. The following is a copy of the survey “question”:

“To: Members, AASHTO Subcommittee on Materials

The National Transportation Safety Board (NTSB) has contacted our office for information from the AASHTO Member Departments regarding the testing of critical systems (or materials) that do not have AASHTO or ASTM test methods or specs associated with them. For example, a system used to support components suspended over a roadway where a material failure would likely cause an accident involving vehicles and/or pedestrians. How does your State ensure that these critical systems or materials will work properly? Please check one or more of the boxes below:

A. The state evaluates the material in-house (or hires a lab to evaluate it) based on a non-AASHTO protocol

B. The state asks the contractor to evaluate the material (or hire a lab to evaluate it) based on a non-AASHTO protocol

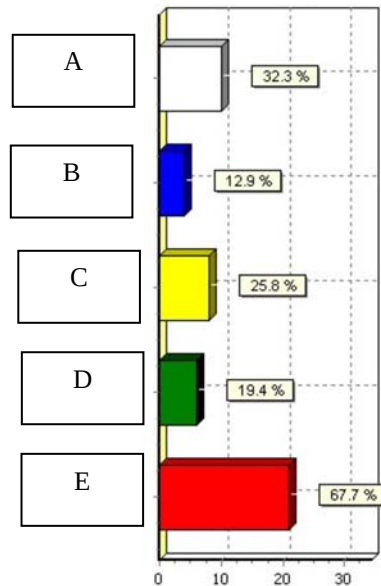
C. The state uses/trusts the manufacturers’ specs or certifications

D. The state delegates responsibility to the contractor to ensure that these critical systems will work properly (the contractor, in turn, may simply use the manufacturers’ specs or certifications)

E. Other (please explain)

Please provide any additional explanation you think would be helpful in understanding how your state handles this situation.”

Thirty states responded to the request by AASHTO with the following results:



Responses in the “Other E” category included:

A qualified products list (QPL) is developed based on a corresponding procedure that includes independent lab testing from the supplier and our lab testing for verification. Contractor performs witnessed proof testing in the field to confirm personnel ability and procedures. Quality control (QC) testing is done on the first day of production installation. Quality assurance (QA) testing is done on a random basis. Criteria and requirements for QC and QA testing are cited in the specifications.

Your survey is not functioning as stated in #3. I was only permitted to select one. Therefore, my selection is actually a, c and other.

The Professional Engineer will have to verify the design and testing to assure the system will meet the critical application.

How handled varies. If proposed on a working contract an ad-hoc group is formed to evaluate/test. Often, we ask for independent lab results from the contractor/vendor, but attempt to reconfirm testing in our own lab. If a vendor submitted item and not time critical, it goes to our Products Evaluation Unit and may be accepted or rejected by the Highway Developmental Council (standing group to evaluate non-standard materials). In both cases we research what would be the most appropriate testing protocols.

When material can not be tested in-house we require notarized test results and certifications from the manufacturer.

We recently completed a Materials Risk Analysis, evaluating all common materials for two kinds of risk: failure to meet specification and consequences if specifications are not met. This analysis provides a basis for how we accept materials: based on testing, fabrication inspection, manufacturer's cert of compliance, visual, etc.

We'd try and perform ourselves or use a consultant services provider to ensure compliance.

Kentucky does have a New Products procedure to address materials that have no existing specification.

MD has a new products committee that uses a standard form to get all pertinent information on the material in question. We would ask the manufacturer of that material to provide contacts, preferably DOT, where the material had been used if possible and provide any test data that they have on the material for our review. Based on this information we would make a decision to proceed with evaluating the material for use on a project specific basis or on a statewide basis.

Additional methods for critical systems or materials that have been utilized include a mandatory system operation, or materials function, failure free time period. During this time the contractor must maintain, rectify, correct any and all operational issues at their cost. The time cannot conclude nor final payment made until the specified time period has passed continuously without issues.

Again it would depend on the material. If we could develop our own test method or specifications to include in the contract we would. If no specifications exist, we may require the supplier to provide specifications, test results, and certifications on the product. In some instances we may accept the product based only on the suppliers certification. It is hard to give a specific answer without knowing of the specific product.

We use LADOTD, AASHTO, & ASTM test procedures, Certificates of Compliance, Analysis, and Delivery. We use a great deal of in-house testing and use some contractor results in the acceptance process.

For critical items as described above, Pennsylvania requires the manufacturer/contractor to provide engineering calculations to support the use of the material/system in question. We may also test some of the materials to verify the properties supplied by the manufacturer/contractor.

We would in some situations evaluate the component itself to determine the appropriate action. It could be a combination of the items in question 3. If at all possible we would try to find a AASHTO or ASTM reference that would at least support the direction of our action.

In some cases we maintain an approved products list that "qualifies" certain materials where we review and test the material or system and then it is available for any project. The answer is further clarified as such - if we do not have the equipment or expertise to test the material in house we would identify outside testing expertise and, dependant on how critical the element is, we may even witness the testing.

We do not "ask" contractor to evaluate critical systems or materials that do not have AASHTO or ASTM specification, but we have "required" the contractor to test items and we witness the testing.

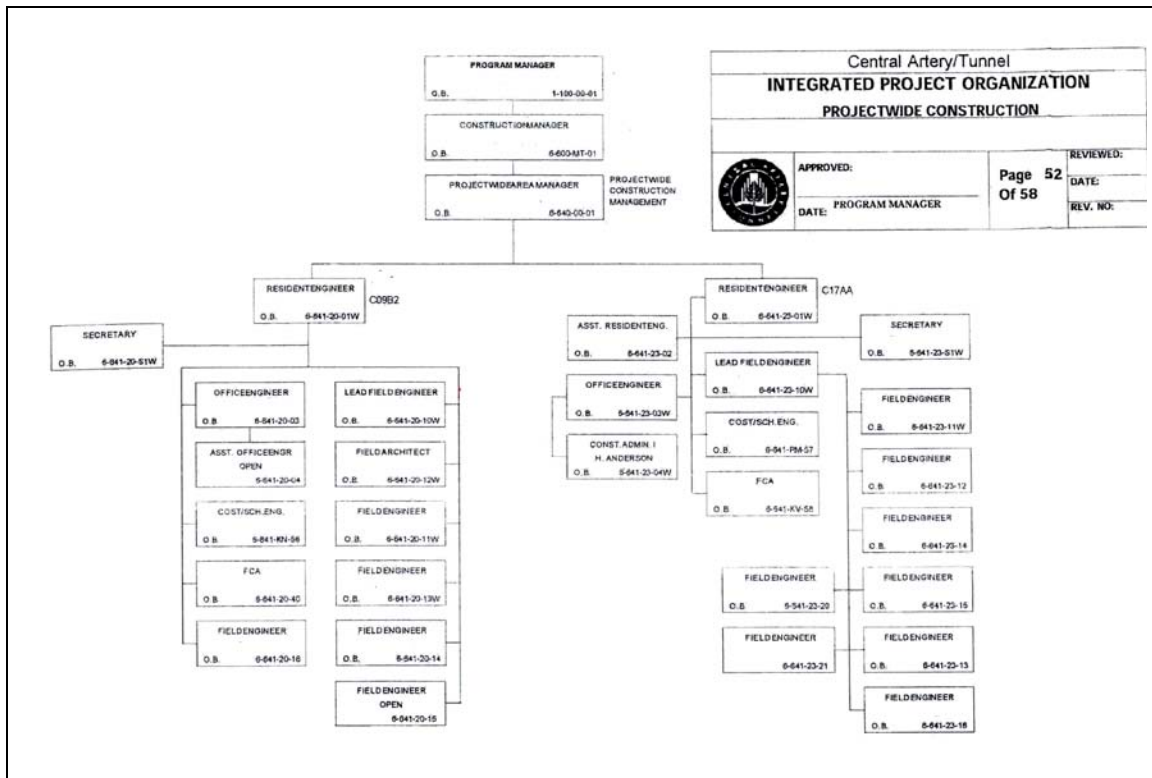
Most structure critical components are tested in our Central Materials Laboratory. Generally, if there is not an accepted AASHTO or ASTM test method, there is a state test method outlining the procedure to be used. Certain components are taken on manufacturers' certification and/or a certified test report.

Require that a registered professional engineer, registered in any state, evaluate and certify that the item design complies with the plans and specifications and meets and exceeds the appropriate national standard and that the materials meet or exceeds the specification requirements. In addition, require the Contractor to furnish a certification that the item is fabricated in compliance with the certified design and complies with the specification requirements. For devices that require crash testing, the manufacturer is required to supply a certification that the device has been crashed tested and meets requirements per the appropriate crash test specified.”

H3. Bechtel/Parsons Brinkerhoff Program Management Project C09B2¹⁰

Documents associated with the B/PB's management of the accident C09B2 project were searched including the Field Engineer's Daily Report (FEDR) system and documents, memos and E-mails. There were 12 engineering personnel positions allocated for the accident project. The highest ranking person was the “Resident Engineer”. This person was responsible for C09B2 only. He reported to the “Project Area Wide Manager”, who in turn reported to the Construction Manager for the entire program. The Construction Manager reported to the Program Manager who was the highest level of the organization.

¹⁰ The “Field Architect” assigned to this C09B2 had previously been assigned to the Ted Williams Tunnel (TWT) project in a function similar to the one he performed on the C09B2. Specifically he kept track of the progress and difficulties with the installation, testing and associated with the adhesive anchors.



C09B2 Project Organization Chart

The FEDR documents for the C09B2 related to the installation of the tunnel ceilings were reviewed and copied for further reference by Board engineers. Several field engineers and a field architect closely monitored the installation of epoxy adhesive anchor bolts for the ceiling panels beginning on or about 2 June 1999. The process extended well into 2000. The bolts were also used for other fastening requirements including numerous wall panels and seismic braces in the tunnel. The B/PB engineers' comments appear in FEDR records almost every day. The field engineers include number of bolts installed and critiques on the process of drilling holes, like failure to properly clean out the drilled holes. These conditions were relayed to the Contractor Modern Continental. Logs were also maintained of the number of holes drilled, bolts installed and pull tested by the materials testing firm Conam, INC. There are also documented showing the location of the bolts and the date installed and test results. The location and number of failed and passed pull tests were also documented.

Early emphasis was given to the HOV lanes where the Contractor was to experiment with the installation of the anchors, assembly of the ceiling panels, and rigging of the ceiling support structure. The area where this process occurred was called the "mock-up area".

On 13 October 1999 an entry in the FEDR¹¹ states that the Field Architect met with personnel from Modern Continental and Powers Fasteners (the supplier of the adhesive epoxy) to discuss a problem with adhesive anchor bolts.

¹¹ FEDR page 71

“ FArch met with MCC and Powers Fasteners, Inc. atop the mock-up ceiling to discuss the problem of anchor bolts "creeping" out over time. We showed them the problem (displaced hanger plates/bolts) and reviewed the anchor bolt installation and testing procedures, and module erection process. We gave them a copy of FArch's 10/08/99 field sketch. The Powers representatives did not offer any analysis or conclusions at this time, preferring to confer with their company's engineering staff before offering opinions and/or solutions. (Ref Dwg s-387; Ref Spec 723.480.3.02; Ref Submtl 723.480-009-A).

On 21 October 1999 the following entry appears in the FEDR¹²:

“-At approximately 10:15AM a field mtg. was held with the Field Engineering Manager for Powers Fasteners, from New Rochelle, NY. along with the following additional personnel
NMCC Senior Project Engineer

MCC QC Manager

MCC Superintendent

MCC Asst. Supt.

B/PB LFE

B/PB Field Architect –

The subject of the meeting was to discuss and determine the ceiling module HOV078. It should be noted that Powers Fasteners is the manufacturer of the epoxy used to install the 5/8" stainless steel ceiling anchors. MCC reviewed with their personnel the installation and testing procedures that were used for installing the ceiling anchors.

Several potential areas of concern were discussed:

1) Installation temperature. It was determined that this was not an issue as the epoxy was installed during the months of July & August when the temperatures in the tunnel were considerably above the minimum required.

2) Reduction in load capacity due to lack of adhesion with cut reinforcing steel. It was discussed that the worse case would be if two #11 bars were penetrated at their intersection. This would reduce the design hole depth to an effective embedment length

of 2.75 inches, which is a 50% reduction of the contact area with concrete. According to the manufacturer's data the capacity of the anchor would then be just under 10.0K rather than the 31.0+K if the full effective depth of 5.5 inches were achieved. However, this approximately 10.0 K is still greater than the 3.25K at which each anchor was pull tested.

3) Lack of adhesion due to using a diamond tipped core drilling bit rather than a carbide bit. It was noted that a carbide bit would give a rougher surface inside the hole and hence greater adhesion, but that it was not possible to use a carbide bit due to the amount of r” (END OF PAGE – NO further text.

On 12 November the following entry appears in the FEDR¹³:

“12 November 1999

Two technicians from Conam Inspection Inc. conducted pull out tests on 2 bolts in WB which were previously skipped, and both bolts passed. The technicians then went to the mock-up area to pull out one of the adhesive anchor bolts which had previously passed the pullout test, but had been slipping out over the last several months.

The bolt had passed the pull-out test on 7/30/99. However, a survey conducted on 10/08/99 found that the bolt had slipped out 9/16", the most displacement of the 5 bolts that were slipping. When the standard pull-out test load (3.25 kips) was applied today, the bolt held for a few seconds, then began to pull out with almost no resistance. The failed bolt was removed and a new bolt will need

¹² FEDR page 76

¹³ FEDR page 96.

to be installed and re-tested. Upon examining the pulled out bolt, it appeared to lack sufficient epoxy to fully fill the drilled hole. Also, there was a significant amount of concrete dust adhered to the epoxy surrounding the bolt, usually an indication that the drilled hole was not completely cleaned out prior to installation. The epoxy nearest to the embedded tip of the bolt was brittle and easily crumbled, usually an indication of improper mixing of resin and hardener. The epoxy near the exposed end of the bolt (but still in the embedded portion) appeared to be fully and properly cured. This would indicate the mixture of resin and hardener was not uniform during the injection of epoxy into the drilled hole. In spite of this combination of installation deficiencies it appears the bolt was able to develop just enough strength to pass the pull-out test. However after several weeks of constant loading by the ceiling module, the bond was broken, and the bolt began to slip out. (Ref D (Ref Dwg S-377; Ref Spec 723.480.2.05; Ref Submtl 723.480-008-A). WBD080A, WBD080B, WBD081A, WBD081A, WBD081B and WBD082B, a total of 5 modules for 2400 sq ft. (Ref Dwg S-269; ceiling modules in WB. By COB they had hung modules Ref Specs 723.480 & 960.020; Ref Submtls 723.480-004-A & 960.020-010-B)”

FEDR entries continue document in the same degree of detail activity involving the installation of epoxy adhesive anchor bolts in the tunnels until the completion of the project. A “service report” (SD) was also initiated by B/PB. This was a formal process that dealt with problems identified by B/PD. It required an investigation and solution of the deficiency before proceeding with construction.

There is other documentation showing P/PB’s interaction with the Section Design Consultant Gannett Fleming regarding the use of epoxy adhesive anchor bolts for the ceiling panels during the design stage of the project. Gannett Fleming recommended the use of “under-cut” expansion bolts. Ultimately B/PB decided upon the use of the epoxy bolts.

H4. Tunnel Inspections

As part of their contract B/PB produced a tunnel inspection manual for the State. The introduction to the manual contains the following information:

“1.1 Purpose and Scope

The purpose of the tunnel inspection manual is to aid the Massachusetts Turnpike Authority (MTA) in the detailed and scheduled inspection of the tunnels, boat sections and the structural elements included within the Central Artery/Tunnel (CA/T) network. Tailored to this highway tunnel system, the scope encompasses all aspects of a structural inspection.”

“2.0 Development of Protocols

2.1 Inspection Type and Frequency

The tunnel systems, boat sections and ventilation building structures inspection frequency has not been determined and will be developed in cooperation with the MTA. The FHWA may require a frequency of every two years, similar to the present State Bridge Inspection Program. The schedule will vary with the condition of the structure. As

a structure becomes deficient, closer intervals may be required e.g. an annual/yearly “visual inspection”. A routine, in-depth inspection should be performed every 3-5 years.”

5.0 General Tunnel Types

Page 40 of the manual provides plan sheet references for the accident tunnel C09B2 and specific directions for its inspections.

5.2.1 Concrete Components

Inspection of exposed concrete elements in the tunnel and boat sections include:

1. See Section 5.1.5 for TWT tube joints J1 - J11 and ITT C09B1 Gina type
2. Roof / Lining Concrete
3. Ceiling Panels and Supports
4. Walls and Wall Panels

The accident tunnel was opened to traffic in 19 Jan 2003. Had it been inspected according to the Tunnel Inspection document’s requirement it would have been inspected at least twice before the accident. Although this area had a history of failed anchor bolts there were no documented inspections in the area of the accident of in the I-90 tunnels before the accident.